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Saveetha Institute of Medical And Technical Sciences
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Security Attacks, Threats, Risks and Their Impact on Security Goals

CSA5116-CRYPTOGRAPHY AND NETWORK SECURITY

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Introduction to Information Security

- **What are Security Goals?**

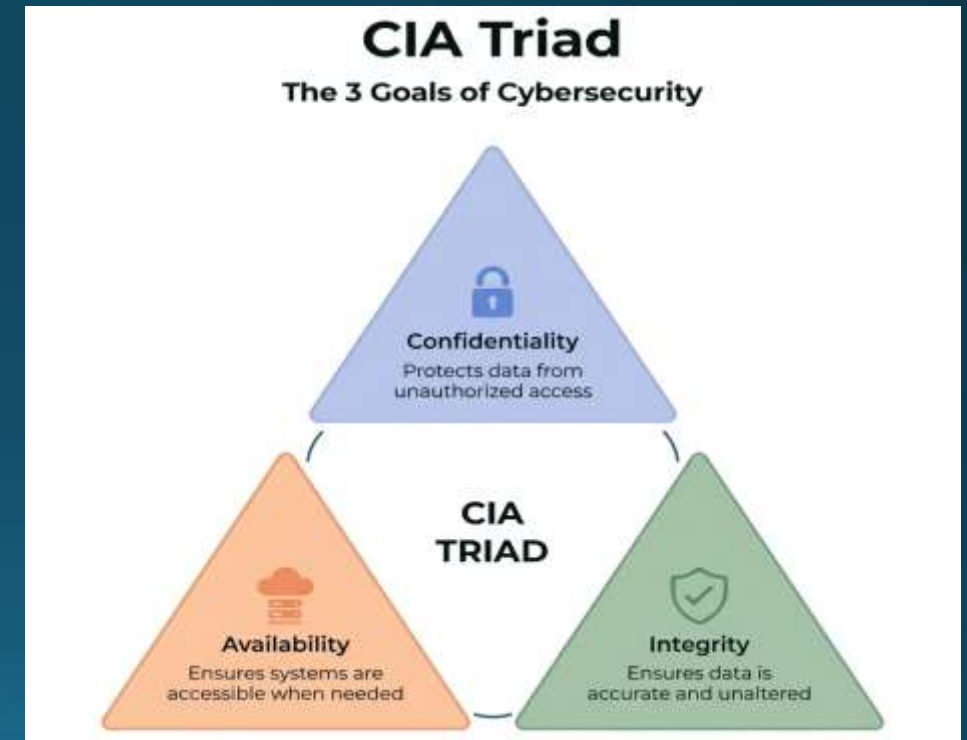
Security goals are the fundamental principles that protect information and ensure secure communication.

- **Why are They Important?**

- Protect sensitive information
- Prevent unauthorized access
- Ensure reliable and secure systems

Core Security Goals

- **Confidentiality** – Protects data from unauthorized access.
- **Integrity** – Ensures data remains accurate and unaltered.
- **Availability** – Ensures authorized users can access information when needed.





Understanding Threats, Attacks & Risks

➤ Threat

- A potential danger that can harm a system or its information.
- **Examples:** Hacker, Malware, Natural Disaster

➤ Attack

- An intentional attempt to exploit a system and gain unauthorized access.
- **Examples:** Phishing, Ransomware, DoS Attack

➤ Risk

- The chance of loss or damage when a threat exploits a vulnerability.
- **Examples:** Data theft, Financial loss, Service disruption

Threat → Vulnerability → Attack → Risk → Impact

- **Hacker (Threat) → Phishing Email (Attack) → Stolen Password (Risk)**

Types of Security Attacks

- **Passive Attacks**
 - *Observe information without altering it.*
 - **Release of Message Contents** – Unauthorized reading of data.
 - **Traffic Analysis** – Monitoring communication patterns.
- **Active Attacks**
 - *Modify, disrupt, or destroy information.*
 - **Masquerade** – Impersonating an authorized user.
 - **Replay Attack** – Resending intercepted data.
 - **Message Modification** – Altering transmitted messages.
 - **Denial of Service (DoS)** – Disrupting access to services.

Passive Attack	Active Attack
▪ Observes data	▪ Modifies or disrupts data
▪ Difficult to detect	▪ Easier to detect
▪ Affects Confidentiality	▪ Affects Integrity & Availability

Security Measures

➤ Encryption

Protects sensitive information by converting it into unreadable cipher text.

➤ Authentication

Verifies the identity of users before allowing access to systems or data.

➤ Access Control

Restricts access to information based on user roles and permissions.

➤ Backup & Recovery

Ensures important data can be restored after cyberattacks or system failures.





Impact of Security Threats & Conclusion

Data Breaches

- Unauthorized access to confidential or sensitive information.

Financial Loss

- Costs incurred due to cyberattacks, fraud, random payments, and system recovery.

Operational Disruption

- Interruptions in business operations caused by attacks such as malware or Denial-of-Service (DoS).

Identity Theft

- Personal or organizational information is stolen and misused for fraudulent activities.